FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUIT

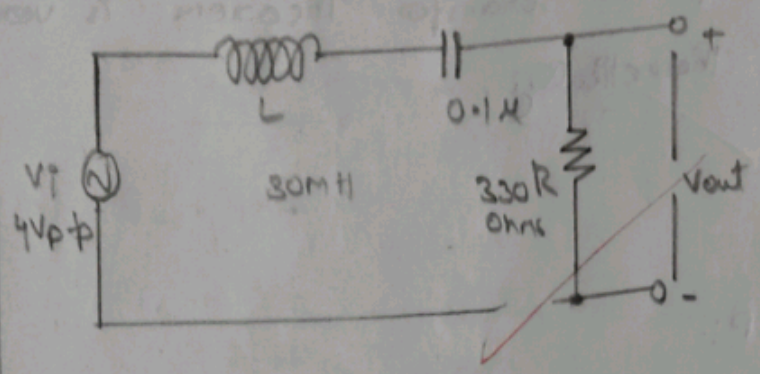
Aim :-

To study the behaviour of series and parallel RLC circuit at resonance and to determine the resonance frequency, bandwidth and Q of the given RLC circuit using Multisim

Apparatus required :-

S.NO	Components	specification	Quantity
1	CRO	10 Mhz	1
2	DIB	–	1
3	capacitors	0.1 μF	1
4	Resistors	330 Ω	1
5	Breadboard	–	–
6	Connecting wires	–	–
7	Pc with Multisim	–	–

CIRCUIT DIAGRAM



PROCEDURE:-

1. Construct circuit shown in figure 1 on breadboard

2. Connect CRO ch 1 to input and ch_2 to output

3. Set the input voltage to $4V_{p-p}$

4. Vary the frequency from 500 Hz to 3 KHz in small steps to get a maximum output voltage magnitude.

The frequency at this maximum voltage V_m is the resonance frequency.

5. If $V_{cutoff} = V_m/\sqrt{2}$, vary the frequency again until the output voltage equals V_{cutoff}.

6. Calculate the bandwidth by subtracting the 2 cut-off frequencies

7. Calculate the Q factor

8. Repeat step 1 through 7 for circuit shown in Figure 2.

TABULATION: $V_{in} = 4V_{P-P}$.

Series Resonance			Parallel Resonance		
Frequency (Hz)	Output Voltage (V)	Output Current (mA)	Frequency (Hz)	Output Voltage (V)	Output Current (mA)
500	1.2	2	500	1.5	9
700	2.0	4	700	2.8	7
900	3.5	7	900	4.0	5
1000	5.0	10	1000	5.0	2
1100	3.6	7	1100	4.1	5

Result :-

Thus, the phenomenon of resonance in RLC circuit was studied and the following were determined using multisim simulator.

Series RLC circuit

Resonant frequency = 1000

Bandwidth = 525.4

Q factor = 1.908

parallel RLC circuit

Resonant freq = 1000

Bandwidth = 525.4

Q factor = 1.908.

S.NO	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1.	EXP SETUP / CONNECTIONS	20	20
2.	OBS / READINGS	20	20
3	Calculation / Result	20	20
4	Viva	20	20
5	Record	20	20
			100

CALCULATION :-

$L = 100mH = 0.1H$

$C = 0.25\mu F = 0.25 \times 10^{-6} F$

$$f_0 = \frac{1}{2\pi\sqrt{(0.1)(0.25 \times 10^{-6})}}$$

$$f_0 = \frac{1}{2\pi\sqrt{1.58 \times 10^{-4}}}$$

$$f_0 = \frac{1}{9.92 \times 10^{-4}}$$

$$f_0 = 1000 Hz$$

Maximum current and minimum current occur at 1000 Hz

$$BW = \frac{R}{2\pi} = \frac{330}{2 \times 3.14 \times 0.1} = \frac{330}{0.628} = 525.4$$

$$Q = \frac{f_0}{BW} = \frac{1000}{525.4} = 1.903$$